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Thermally assisted magnetic head, head gimbal assembly, and hard disk drive

Abstract

The disclosure provides a thermally assisted magnetic head comprising a slider and a light source unit provided on the slider. An intermediate connection pad is provided in such a manner that the first light source connection pad and the second light source connection pad of the slider are arranged in a row on a straight line while the intermediate connection pad is not located on the said straight line. The first electrode of the light source unit is electrically connected with the first light source connection pad, and both the second electrode of the light source unit and the second light source connection pad are electrically connected with the intermediate connection pad. Accordingly, a head gimbal assembly and a hard disk drive are further provided in the embodiment of the disclosure.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6404706	12/2001	Stovall	N/A	G11B 5/00
7898909	12/2010	Shimazawa et al.	N/A	N/A
8116173	12/2011	Shimazawa et al.	N/A	N/A
8134794	12/2011	Wang	369/13.13	G11B 5/314
8194509	12/2011	Shimazawa et al.	N/A	N/A
8223596	12/2011	Chou et al.	N/A	N/A
8325566	12/2011	Shimazawa et al.	N/A	N/A
8391106	12/2012	Shimazawa et al.	N/A	N/A
10566013	12/2019	Honda	N/A	G11B 5/012
2010/0195239	12/2009	Takayama	N/A	N/A
2011/0075527	12/2010	Hirata	N/A	G11B 7/1387
2011/0317527	12/2010	Wang	369/13.02	G11B 5/127
2012/0044790	12/2011	Shimazawa et al.	N/A	N/A
2024/0321299	12/2023	Aoki	N/A	G11B 5/4826

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102254839	12/2010	CN	N/A
102376313	12/2011	CN	N/A
115472586	12/2021	CN	N/A

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Background/Summary

(1) This application is claims priority to CN 202410216890.3 filed Feb. 27, 2024, the entire contents of which is hereby incorporated by reference.

TECHNICAL FIELD

(2) The present disclosure relates to the technical field of hard disk drives, more particularly to a thermally assisted magnetic head, a head gimbal assembly, and a hard disk drive.

BACKGROUND

(3) The hard disk drive (HDD) is used for writing/reading data to/from a disk. As shown in FIG. 1, a hard disk drive **300** generally comprises a plurality of rotatable disks **301** connected to a spindle motor **302**, a drive arm **304**, and a head gimbal assembly (HGA) **200** mounted at an end of the drive arm **304**. Typically, the hard disk drive **300** may further include a voice-coil motor (VCM) to control the movement of the driving arm **304**. The head gimbal assembly **200** comprises a magnetic

head **230** and a suspension **290** for supporting the magnetic head **230**. When the hard disk drive **300** works, the disk **301** is rotated at a high speed by the spindle motor **302**, and the magnetic head **230** flies above the disk due to the air pressure generated by the rotation of the disk **301**. Under the control of the voice-coil motor, the magnetic head **230** moves in a radial direction on the surface of the disk **301**. Concerning different tracks, the magnetic head **230** can read/write data from/to the disk **301**.

(4) In the technology of heat assisted magnetic recording (HAMR), the magnetic head comprises a slider and a light source unit (such as a laser diode) arranged on the slider. The light source provides energy to a part of the magnetic recording medium so as to reduce the coercive force thereof. Furthermore, a write magnetic field can be applied to the part that is heated, to facilitate the write operation.

(5) Now, the slider is provided with a plurality of connection pads arranged in a row, while the suspension is provided with a plurality of corresponding electrically connecting terminals, so as to allow input of control signals to the magnetic head. Herein, the plurality of connection pads arranged in a row on the slider comprise an LDP pad connected to the P-electrode of the laser diode and an LDG pad connected to the N-electrode of the laser diode. For example, referring to FIG. 2 in related art, when the N-electrode of the laser diode is connected to the LDG pad on the slider, the joint where the N-electrode of the laser diode is bonded to the LDG pad and the joint where the corresponding electrically connecting terminal of the suspension is bonded to the LDG pad may be overlapped, which may result in unstable bonding and thus low yield of the magnetic head. To solve the above-described problems, referring to FIG. 3 in related art, the number of connection pads arranged in the row on the slider is reduced, to leave more space for the LDG pad. Namely, the area of the LDG pad is increased. In such a case, the overlap between the joint where the N-electrode of the laser diode is bonded to the LDG pad and the joint where the corresponding electrically connecting terminal of the suspension is bonded to the LDG pad can be avoided. However, as the plurality of connection pads arranged in the row on the slider provide various functions, such as readers, writers, and HDI sensors, in addition to the controlling of the laser diode, the method in which the number of connection pads arranged in the row on the slider is reduced cannot meet multiple performance requirements for the magnetic head.

SUMMARY

(6) Aiming at the shortcomings of existing technologies, the embodiment of the disclosure provides a thermally assisted magnetic head, which comprises a slider and a light source unit arranged on the slider;

(7) The slider is provided with a first light source connection pad, a second light source connection pad, and an intermediate connection pad, wherein the first light source connection pad and the second light source connection pad are arranged in a row on a straight line while the intermediate connection pad is not located on the said straight line; The light source unit is provided with a first electrode and a second electrode, wherein the first electrode is electrically connected with the first light source connection pad, the second electrode is electrically connected with the intermediate connection pad, and the second light source connection pad is electrically connected with the intermediate connection pad.

(8) Preferably, the intermediate connection pad may be a connection pad exclusively electrically connected with the second electrode and the second light source connection pad.

(9) Preferably, the intermediate connection pad and the light source unit may be arranged in a row.

(10) Preferably, the intermediate connection pad may be an electrical lapping guide connection pad.

(11) Preferably, the second electrode and the intermediate connection pad may be joined by wire connection or solder ball connection.

(12) Preferably, the solder ball connection may be achieved by SBB bridge welding.

(13) Preferably, the slider may be provided with a mounting surface, and all of the first light source connection pad, the second light source connection pad, and the intermediate connection pad are

disposed on the mounting surface; the first electrode may be disposed on the first surface of the light source unit, the second electrode may be disposed on the second surface of the light source unit, the first surface is opposite to the second surface and is located at the mounting surface.

(14) Preferably, the light source unit may be a laser diode, the first electrode may be one of the P-electrode of the laser diode and the N-electrode of the laser diode, the first light source connection pad may be the P-electrode connection pad or N-electrode connection pad correspondingly, the second electrode may be the other one of the P-electrode of the laser diode and the N-electrode of the laser diode, and the second light source connection pad may be the P-electrode connection pad or N-electrode connection pad correspondingly.

(15) In order to solve the above-mentioned technical problems, a head gimbal assembly is further provided in the embodiment of the disclosure, which comprises the above-mentioned thermally assisted magnetic head.

(16) In order to solve the above-mentioned technical problems, a hard disk drive is further provided in the embodiment of the disclosure, which comprises the above-mentioned head gimbal assembly.

(17) Compared with existing technologies, the embodiment of the disclosure has advantages as follows. In the embodiment of the disclosure where the thermally assisted magnetic head comprises the slider and the light source unit provided on the slider, the intermediate connection pad is provided in such a manner that the first light source connection pad of the slider and the second light source connection pad of the slider are arranged in a row on the straight line while the intermediate connection pad is not located on the said straight line. As the first electrode of the light source unit is electrically connected with the first light source connection pad, and both the second electrode of the light source unit and the second light source connection pad are electrically connected with the intermediate connection pad. Thus, a reliable electrical connection between the second electrode of the light source unit and the second light source connection pad can be ensured. In this way, it avoids the low yield caused by the overlap of the joint where the second electrode of the light source unit is bonded to the second light source connection pad and the joint where the suspension is bonded to the second light source connection pad in practice, without the need to reduce original quantity of connection pads. Accordingly, in the embodiment of the disclosure, a head gimbal assembly and a hard disk drive are further provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a structural schematic view of a hard disk drive in related art;
- (2) FIG. 2 is a schematic view illustrating a connecting manner between a laser diode and connection pads of a slider in related art;
- (3) FIG. 3 is a schematic view illustrating a further connecting manner between a laser diode and connection pads of a slider in related art;
- (4) FIG. 4 is a structural schematic view of a thermally assisted magnetic head according to an embodiment of the disclosure;
- (5) FIG. 5 is a left view corresponding to FIG. 4;
- (6) FIG. 6 is a schematic view illustrating an assembling of a thermally assisted magnetic head and a suspension according to an embodiment of the disclosure;
- (7) FIG. 7 is a left view corresponding to FIG. 6;
- (8) FIG. 8 is a schematic view illustrating that a second electrode of a light source unit is welded to an intermediate connection pad via SBB bridge welding;
- (9) FIG. 9 is a left view corresponding to FIG. 8;
- (10) FIG. 10 is a schematic view illustrating an assembling of a thermally assisted magnetic head and a suspension according to an embodiment of the disclosure;

- (11) FIG. 11 is a left view corresponding to FIG. 10;
- (12) FIG. 12 is a schematic view illustrating that an intermediate connection pad is located on a left side of a light source unit according to an embodiment of the disclosure;
- (13) FIG. 13 is a bottom view corresponding to FIG. 12;
- (14) FIG. 14 is a structural schematic view of a thermally assisted magnetic head according to an embodiment of the disclosure, where an intermediate connection pad is an electrical lapping guide connection pad;
- (15) FIG. 15 is a left view corresponding to FIG. 14;
- (16) FIG. 16 is a bottom view corresponding to FIG. 14;
- (17) FIG. 17 is a structural schematic view of an LOS structure in related art;
- (18) FIG. 18 is a schematic view of an LIT structure according to an embodiment of the disclosure;
- (19) Herein, **300.** hard disk drive; **302.** spindle motor; **301.** disk; **304.** drive arm; **200.** head gimbal assembly; **230.** magnetic head; **290.** suspension; **1.** thermally assisted magnetic head; **2.** slider; **21.** first light source connection pad; **22.** second light source connection pad; **23.** intermediate connection pad; **24.** mounting surface; **25.** leading surface; **26.** air bearing surface; **3.** light source unit; **31.** first electrode; **32.** second electrode; **4.** suspension; **5.** waveguide; **6.** surface plasmon antenna; **10.** slider; **20.** laser diode; **30.** base; **40.** surface plasmon antenna; **50.** waveguide.

DETAILED DESCRIPTION OF ILLUSTRATED EMBODIMENTS

- (20) The technical solutions according to the embodiments of the present disclosure will be clearly and completely explained below in conjunction with the drawings for the embodiments. Apparently, the described embodiments are merely a part rather than all of the embodiments of the present disclosure. All embodiments obtained by those skilled in the art based on the embodiments of the present disclosure without creative work shall fall within the scope of the present disclosure.
- (21) As shown in the structural schematic views of a thermally assisted magnetic head **1** according to an embodiment of the disclosure in FIGS. 4-7, a thermally assisted magnetic head **1** according to the embodiment of the disclosure comprises a slider **2** and a light source unit **3** provided on the slider **2**;
- (22) The slider **2** is provided with a first light source connection pad **21**, a second light source connection pad **22**, and an intermediate connection pad **23**, wherein the first light source connection pad **21** and the second light source connection pad **22** are arranged in a row on a straight line while the intermediate connection pad **23** is not located on the said straight line; The light source unit **3** is provided with a first electrode **31** and a second electrode **32**, wherein the first electrode **31** is electrically connected with the first light source connection pad **21**, the second electrode **32** is electrically connected with the intermediate connection pad **23**, and the second light source connection pad **22** is electrically connected with the intermediate connection pad **23**.
- (23) In the embodiment of the disclosure where the thermally assisted magnetic head **1** comprises the slider **2** and the light source unit **3** provided on the slider **2**, the intermediate connection pad **23** is provided in such a manner that the first light source connection pad **21** of the slider **2** and the second light source connection pad **22** of the slider **2** are arranged in a row on the straight line while the intermediate connection pad **23** is not located on the said straight line. As the first electrode **31** of the light source unit **3** is electrically connected with the first light source connection pad **21**, and both the second electrode **32** of the light source unit **3** and the second light source connection pad **22** are electrically connected with the intermediate connection pad **23**. Thus, a reliable electrical connection between the second electrode **32** of the light source unit **3** and the second light source connection pad **22** can be ensured. In this way, it avoids the low yield caused by the overlap of the joint where the second electrode **32** of the light source unit **3** is bonded to the second light source connection pad **22** and the joint where the suspension **4** is bonded to the second light source connection pad **22** in practice, without the need to reduce original quantity of connection pads.
- (24) Referring to FIGS. 4, 8, and 12, in an optional example, the intermediate connection pad **23** is

exclusively electrically connected with the second electrode **32** and the second light source connection pad **22**. In the present embodiment, by providing an intermediate connection pad **23** which is connected with both the second electrode **32** of the light source unit **3** and the second light source connection pad **22**, in addition to providing preexisting connection pads, it can avoid the low yield problem caused by the overlap of the joint where the second electrode **32** of the light source unit **3** is bonded to the second light source connection pad **22** and the joint where the suspension **4** is bonded to the second light source connection pad **22** in practice, on the premise of ensuring a reliable electrical connection between the second electrode **32** of the light source unit **3** and the second light source connection pad **22**, without the need to reduce the quantity of preexisting connection pads.

(25) Referring to FIGS. **4**, **6**, **8**, **10**, **12** and **14**, in a particular implementation, the electrical connection between the intermediate connection pad **23** and the second light source connection pad **22** can be achieved for example through printed circuits or the like, and the electrical connection between the first electrode **31** of the light source unit **3** and the first light source connection pad **21** can be achieved for example through printed circuits or the like.

(26) Referring to FIGS. **6**, **7**, **12** and **13**, the intermediate connection pad **23** and the light source unit **3** are arranged in a row. Illustratively, the intermediate connection pad **23** for example may be provided on the left side of the light source unit **3** as shown in FIGS. **12** and **13**, or alternatively may be provided on the right side of the light source unit **3** as shown in FIGS. **6** and **7**, to facilitate the connection between the second electrode **32** of the light source unit **3** and the intermediate connection pad **23**. Consequently, it can improve convenience in actual operation and meanwhile utilize the space of the slider **2** more efficiently.

(27) Referring to FIGS. **14-16**, in a further optional example, the intermediate connection pad **23** may be the electrical lapping guide connection pad. It should be noted that the electrical lapping guide (ELG) connection pad is mainly applied to the lapping process for the slider **2**. When the thermally assisted magnetic head **1** is actually used, the ELG connection pad is not electrically connected with any other element. Thus, the electrical lapping guide connection pad can be used as the intermediate connection pad **23** without affecting any other function of the thermally assisted magnetic head **1**.

(28) Referring to FIG. **4**, in an optional example, the second electrode **32** may be joined with the intermediate connection pad **23** through a wire. Apparently, the second electrode **32** and the intermediate connection pad **23** may be connected by solder ball bonding (SBB) as shown in FIG. **8**. In the present embodiment, the wire connection or the solder ball connection can ensure a reliable connection between the second electrode **32** of the light source unit **3** and the intermediate connection pad **23**.

(29) Referring to FIGS. **8-11**, in an optional example, the solder ball connection may be achieved by SBB bridge welding. In the present embodiment, as the electrical connection between the second electrode **32** of the light source unit **3** and the intermediate connection pad **23** is achieved by SBB bridge welding, the reliability of the connection between the second electrode **32** of the light source unit **3** and the intermediate connection pad **23** can be further ensured.

(30) Referring to FIGS. **8** and **9**, in an optional example, the slider **2** may be provided with a mounting surface **24**, and all of the first light source connection pad **21**, the second light source connection pad **22** and the intermediate connection pad **23** may be disposed on the mounting surface **24**; wherein the first electrode **31** may be disposed on the first surface of the light source unit **3**, the second electrode **32** may be disposed on the second surface of the light source unit **3**, the first surface is opposite to the second surface and is located at the mounting surface **24**.

(31) In the present embodiment, all of the first light source connection pad **21**, the second light source connection pad **22**, the intermediate connection pad **23**, and the light source unit **3** may be arranged on a same surface of the slider **2**, thereby facilitating more compact layout.

(32) In an optional example, the light source unit **3** may be a laser diode. Referring to FIG. **17**, the

LOS (LD On Slider) structure is one of the conventional HAMR structures (i.e., the structures where the laser diode **20** is disposed on the slider **10**), which involves three elements, i.e., the laser diode **20**, the base **30** and the slider **10**. Herein, the laser diode **20** is connected to the base **30**. Such structure is referred to as LDU. The LDU is optically aligned and is connected to the back surface of the slider **10**. Usually, such assembly is referred to as the L-slider (L-SLD). However, the HAMR structure in the present embodiment is LIT (LD In Trailing surface) structure, that is, the structure where the laser diode is arranged in the trailing surface of the slider **2**. In particular, in the LIT structure, the trailing surface of the slider **2** may be provided with a cavity, in which the laser diode is placed. Such design which combines the laser diode in the cavity does not need a specialized laser diode base or even further a connection device thereof, and thus it can reduce material cost and process cost. Furthermore, the LIT structure can contribute to the multi-stack HDD. Referring to FIG. **1**, the mounting surface **24** in the present embodiment is the trailing surface of the slider **2**. Besides, the slider **2** has a leading surface **25** and an air bearing surface (ABS) **26**, and the leading surface **25** is opposite to the trailing surface, as shown in FIG. **9**.

(33) Referring to FIGS. **9** and **18**, in practical application, the air bearing surface **26** of the slider **2** may face the magnetic recording medium, the laser emitted from the laser diode can be transmitted by the waveguide **5** and transduced into the near-field light by the surface plasmon antenna **6**, so as to heat the magnetic recording medium and reduce the coercive force of the magnetic recording medium to facilitate data writing on the magnetic recording medium by the magnetic head.

(34) In a particular implementation, the first electrode **31** may be one of the P-electrode of the laser diode and the N-electrode of the laser diode, the first light source connection pad **21** may be the corresponding P-electrode connection pad or N-electrode connection pad, the second electrode **32** may be the other one of the P-electrode of the laser diode and the N-electrode of the laser diode, and the second light source connection pad **22** may be the corresponding P-electrode connection pad or N-electrode connection pad.

(35) Illustratively, in the case that the first electrode **31** is the P-electrode of the laser diode, the first light source connection pad **21** correspondingly is the P-electrode connection pad, the second electrode **32** the N-electrode of the laser diode, and the second light source connection pad **22** correspondingly is the N-electrode connection pad; in the case that the first electrode **31** is the N-electrode of the laser diode, the first light source connection pad **21** correspondingly is the N-electrode connection pad, the second electrode **32** is the P-electrode of the laser diode, and the second light source connection pad **22** correspondingly is the P-electrode connection pad.

(36) It should be noted that, in a particular implementation as shown in FIGS. **14-16**, the slider **2** may be further provided with a waveguide **5**, the light emitted from the light source unit **3** may be transmitted by waveguide **5**. More detailed descriptions can be referred to relevant technologies and thus are not repeated herein.

(37) Accordingly, a head gimbal assembly **200** is further provided in an embodiment of the disclosure. Referring to FIG. **1**, the head gimbal assembly **200** comprises the thermally assisted magnetic head according to any one of the above described embodiment. In a particular implementation, the head gimbal assembly may further comprise a suspension **290** for supporting the magnetic head. The suspension **290** may be provided with a plurality of electrical connecting terminals, and the slider **2** may be provided with a plurality of connection pads for electrically connecting with the plurality of electrical connecting terminals of the suspension. In this way, control signals can be transmitted by the control circuit through the electrical connecting terminals of the suspension and the connection pads of the slider **2**.

(38) Accordingly, referring to FIG. **1**, a hard disk drive **300** is further provided in an embodiment of the disclosure, which comprises the head gimbal assembly **200**. In a particular implementation, the hard disk drive **300** may further comprise a plurality of rotatable disks **301** connected with the spindle motor **302** and a drive arm **304**, and the head gimbal assembly **200** is mounted at an end of the drive arm **304**. Generally, the hard disk drive **300** further comprises a voice-coil motor for

control the movement of the drive arm **304**. When the hard disk drive **300** works, the disks **301** are rotated at a high speed by the spindle motor **302**, and the magnetic head **230** flies above the disks due to the air pressure generated by the rotation of the disks **301**. Under the control of the voice-coil motor, the magnetic head **230** moves in a radial direction on the surface of the disks **301**. Concerning different tracks, the magnetic head **230** can read/write data from/to the disks **301**.

(39) Compared with existing technologies, the embodiment of the disclosure has advantages as follows. In the embodiment of the disclosure where the thermally assisted magnetic head **1** comprises the slider **2** and the light source unit **3** provided on the slider **2**, the intermediate connection pad **23** is provided in such a manner that the first light source connection pad **21** of the slider **2** and the second light source connection pad **22** of the slider **2** are arranged in a row on the straight line while the intermediate connection pad **23** is not located on the said straight line. As the first electrode **31** of the light source unit **3** is electrically connected with the first light source connection pad **21**, and both the second electrode **32** of the light source unit **3** and the second light source connection pad **22** are electrically connected with the intermediate connection pad **23**. Thus, a reliable electrical connection between the second electrode **32** of the light source unit **3** and the second light source connection pad **22** can be ensured. In this way, it avoids the low yield especially on SBB welding, caused by the overlap of the joint where the second electrode **32** of the light source unit **3** is bonded to the second light source connection pad **22** and the joint where the suspension **4** is bonded to the second light source connection pad **22** in practice, without the need to reduce original quantity of connection pads. Accordingly, in the embodiment of the disclosure, a head gimbal assembly **200** and a hard disk drive **300** are further provided.

(40) All the above are merely preferred embodiments of the present disclosure. It should be noted that those skilled in the art may obtain modifications and equivalents without departing from the technical principle of the present disclosure. The invention is intended to cover all modifications and equivalents included within the scope of the present disclosure.

Claims

1. A thermally assisted magnetic head, comprising a slider and a light source unit arranged on the slider; wherein the slider is provided with a first light source connection pad, a second light source connection pad, and an intermediate connection pad, wherein the first light source connection pad and the second light source connection pad are arranged in a row on a straight line while the intermediate connection pad is not located on said straight line; the light source unit is provided with a first electrode and a second electrode, wherein the first electrode is electrically connected with the first light source connection pad, the second electrode is electrically connected with the intermediate connection pad, and the second light source connection pad is electrically connected with the intermediate connection pad.
2. The thermally assisted magnetic head according to claim 1, wherein the intermediate connection pad is a connection pad exclusively electrically connected with the second electrode and the second light source connection pad.
3. The thermally assisted magnetic head according to claim 2, wherein the intermediate connection pad and the light source unit are arranged in a row.
4. The thermally assisted magnetic head according to claim 1, wherein the intermediate connection pad is an electrical lapping guide connection pad.
5. The thermally assisted magnetic head according to claim 1, wherein the second electrode and the intermediate connection pad are joined by wire connection or solder ball connection.
6. The thermally assisted magnetic head according to claim 5, wherein the solder ball connection is achieved by SBB bridge welding.
7. The thermally assisted magnetic head according to claim 1, wherein the slider is provided with a mounting surface, and all of the first light source connection pad, the second light source

connection pad, and the intermediate connection pad are disposed on the mounting surface; the first electrode is disposed on a first surface of the light source unit, the second electrode is disposed on a second surface of the light source unit, the first surface is opposite to the second surface, and the first surface is located at the mounting surface.

8. The thermally assisted magnetic head according to claim 1, wherein the light source unit is a laser diode, the first electrode is one of a P-electrode of the laser diode and an N-electrode of the laser diode, the first light source connection pad is a P-electrode connection pad or N-electrode connection pad correspondingly, the second electrode is the other one of the P-electrode of the laser diode and the N-electrode of the laser diode, and the second light source connection pad is the P-electrode connection pad or N-electrode connection pad correspondingly.

9. A head gimbal assembly, comprising the thermally assisted magnetic head according to claim 1.

10. A hard disk drive, comprising the head gimbal assembly according to claim 9.

11. A hard disk drive, comprising the thermally assisted magnetic head according to claim 2.

12. A hard disk drive, comprising the thermally assisted magnetic head according to claim 3.

13. A hard disk drive, comprising the thermally assisted magnetic head according to claim 4.

14. A hard disk drive, comprising the thermally assisted magnetic head according to claim 5.

15. A hard disk drive, comprising the thermally assisted magnetic head according to claim 6.

16. A hard disk drive, comprising the thermally assisted magnetic head according to claim 7.

17. A hard disk drive, comprising the thermally assisted magnetic head according to claim 8.
